

DC Motor Angle Control Laboratory

Proportional and Phase Lead Compensator Design

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Introduction

Motor control is a foundational aspect of modern automation, robotics, and electromechanical systems. Precise control over position, speed, and torque enables reliable operation in applications ranging from industrial actuators to medical devices and autonomous vehicles. Achieving accurate and responsive motor behavior is essential for system stability, energy efficiency, and task execution.

This experiment focuses on position control of a DC motor with an inertial load, using both proportional control and a phase-lead compensator. With the simple analytical design of each compensator, the objectives of the laboratory are to:

1. Develop a transfer function model for the DC motor and inertial load system.
2. Design a phase-lead controller that meet performance criteria such as settling time, overshoot, and current limits.
3. Implement both controllers on a National Instruments myRIO and assess their effect on the motor's output angle behavior.
4. Compare the experimental results with MATLAB simulation predictions.

The analytical design of each controller begins with modeling the DC motor and inertial load as a second-order linear system. For the proportional controller, the closed-loop transfer function is derived and analyzed to determine the gain that satisfies time-domain specifications such as settling time and overshoot. For the phase-lead compensator, frequency-domain techniques are used: the desired phase margin and crossover frequency are translated into compensator parameters using Bode plot shaping. Both designs are validated through MATLAB simulations before hardware implementation. Ultimately, mathematical plots for the system's output angle response created in MATLAB will be compared to the actual motion output plots from the motor's behaviour under the constraints of the assigned controllers.

1 Experimental Setup

The experimental setup consists of a direct current (DC) power supply connected to a Lab-View monitor, a Phase-Lead controller and a simple motor with an inertial mass connected to its shaft. The setup is depicted below:

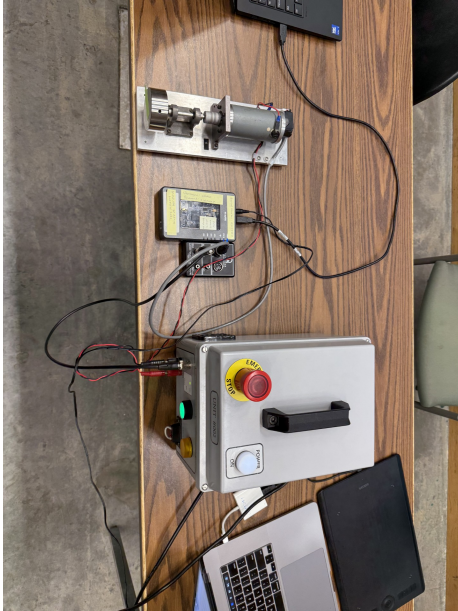


Figure 1: Full Motor Control Physical Setup

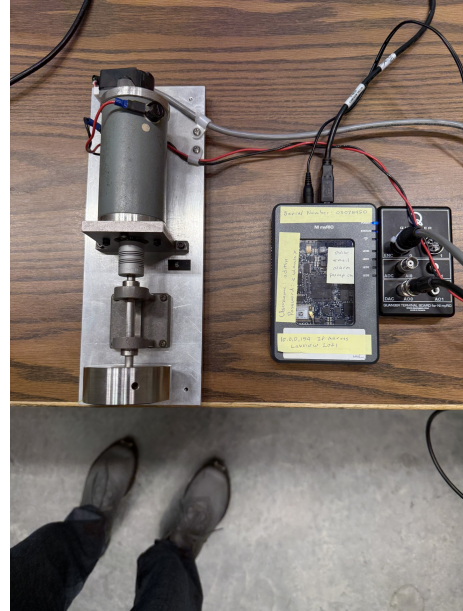


Figure 2: Motor and Controller Setup

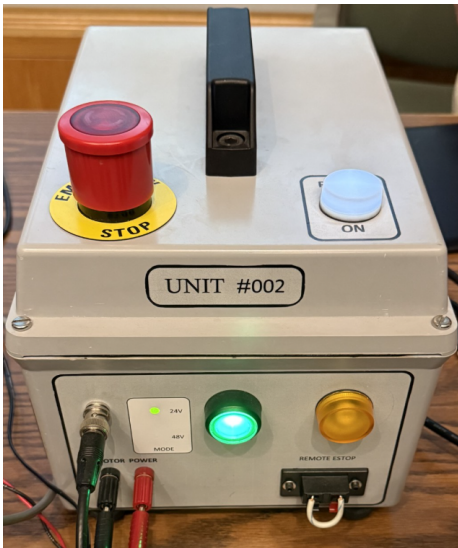


Figure 3; Power Supply Box

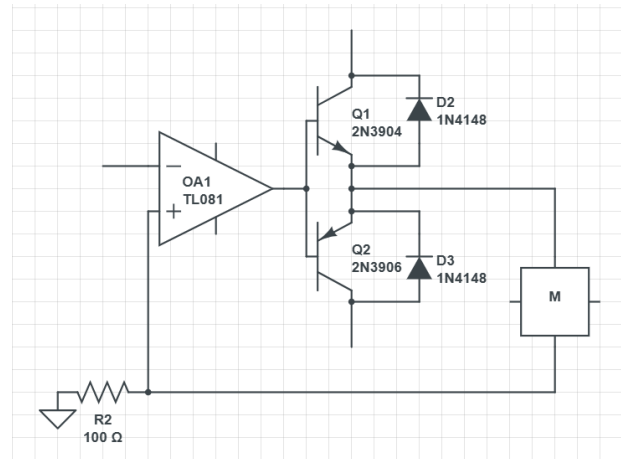


Figure 4: Motor Control Electrical Circuit

2 Plant Model Derivation

The DC motor drives an inertial load consisting of rotor inertia J_M and load inertia J_L , with viscous damping B . The motor torque is proportional to current:

$$T(t) = K_m i(t),$$

and the current is generated from the D/A command voltage $V_{D/A}$ through the amplifier:

$$i_s(t) = K_a V_{D/A}(t).$$

Below is a simplified diagram of the motor model, followed by a linear graph representation of the simplified system. The Linear graph effectively serves as an electrical circuit model for the electromechanical system, showing the series/parallel relationship between components. In the graph, there are two connected systems - one electrical circuit connected to the rotational mechanical output of the motor. Together, these systems form the motor transformer, and they are related to each other through the motor constant.

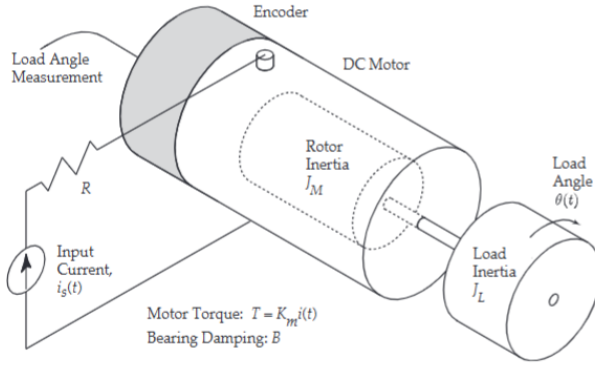


Figure 5: Current-Driven Motor Function Diagram

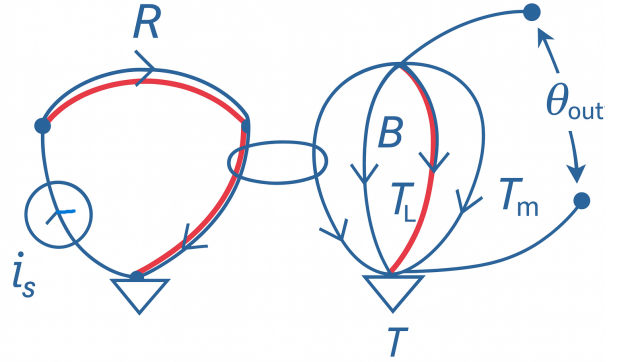


Figure 6: Current-Driven Motor Linear Graph ($T \sim J$)

From the linear graph, it is known that the load inertial mass angular velocity is the system's output state. The transfer function for the output angle with respect to the input current can be derived by treating the system as an electrical circuit, and reducing the circuit to a Thevenin or Norton equivalent to find the input and output impedances.

The equivalent impedance (z_1) of the circuit to the right of the transformer is can be found by adding the impedances of the inertial loads and the dampening:

$$z_1 = \frac{z_J z_B}{z_J + z_B} = \frac{\frac{1}{(J_L + J_m)s} \frac{1}{B}}{\frac{1}{(J_L + J_m)s} + \frac{1}{B}} = \frac{1}{B(J_L + J_m)s \left[\frac{B + (J_L + J_m)s}{B(J_L + J_m)s} \right]} = \frac{1}{B + (J_L + J_m)s}$$

The effective linear graph or circuit is now reduced to the electrical circuit, the transformer and an impedance, as shown:

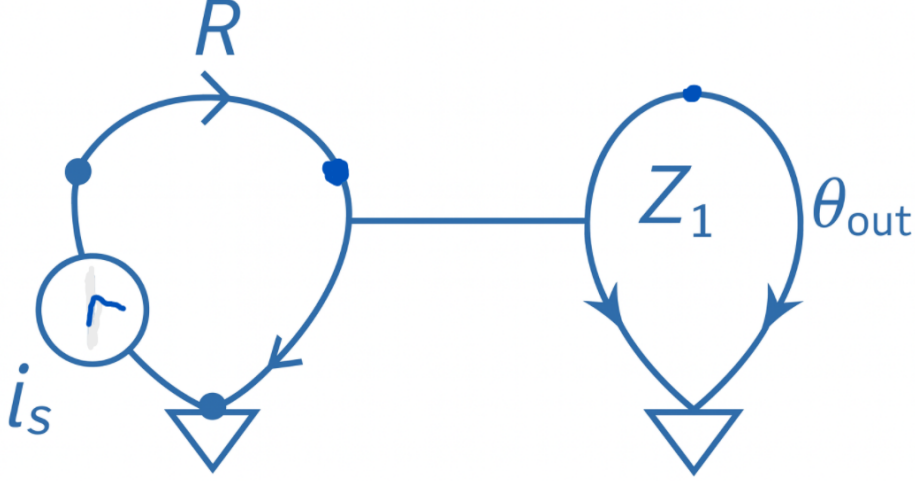


Figure 7: Linear Graph with Reduced Mechanical Port

The equivalent impedance for a linear motor transformer with motor constant defined by $T = K_m i(t)$, in parallel with a single impedance is $z_T = (-K_m)^2 z_1$. Using this relationship, the equivalent impedance for the system's transformer and mechanical side is:

$$z_T = (-K_m)^2 z_1 = (K_m)^2 \frac{1}{B + (J_L + J_m)s} = \frac{K_m^2}{B + (J_L + J_m)s}$$

The Norton equivalent linear graph for the system then becomes the current source, the armature resistance and the equivalent impedance all in parallel, as shown in the final reduced linear graph to the right:

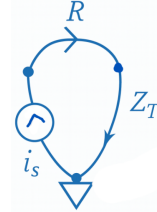


Figure 8: Full Reduced Linear Graph

From this, the transfer function for the output angular velocity with respect to the input current can be written as the output path impedance over the equivalent impedance of the input current. Assuming ideal current gain (no armature resistance), the transfer function for the output angular velocity to the input current is formulated as follows:

$$T_1(s) = \frac{\Omega_L(s)}{I_s(s)} = \frac{-\frac{1}{K_m} V_m(s)}{I_s(s)} = -\frac{1}{K_m} z_T = -\frac{K_m}{B + (J_L + J_m)s}$$

Using the relationship between angular velocity and angle in s ($\Omega(s) = s\theta(s)$) and the relationship between the input current and the D/A voltage ($I_s = K_a V_{D/A}$), the transfer function relating output inertia angle to D/A voltage is:

$$T_m(s) = \frac{\theta_L}{V_{D/A}} = \frac{\frac{1}{s} \Omega_L(s)}{\frac{1}{K_a} I_s(s)} = \frac{K_a}{s} T_1(s) = -\frac{K_a K_m}{s(B + (J_L + J_m)s)} = -\frac{\frac{K_a \cdot K_m}{B}}{s(\frac{(J_L + J_m)}{B}s + 1)}$$

The negative is inherently only there due to the chosen transformer relationships, so it may be ignored for the purpose of controller design. Upon reorganizing the transfer function, a time constant and a D/A gain for the plant can be assigned:

$$T_m(s) = \frac{K_{D/A}}{s(\tau_m s + 1)} \quad \text{where} \quad K_{D/A} = \frac{K_a \cdot K_m}{B} \quad \&\& \quad \tau_m = \frac{(J_L + J_m)}{B}$$

Using the system parameters from the lab handout, numerical values can be obtained:

$$\begin{array}{lll} K_a = 0.41 & B = 0.001 & K_{D/A} = 45.1 \\ K_m = 0.11 & J_M + J_L = 0.00034 & \tau_m = 0.34 \end{array}$$

Final Plant Model:

$$T_m(s) = \frac{45.1}{s(0.34s + 1)}$$

3 Motor Controller Design

The goal of this laboratory is to design two types of controllers for the motor inertia angle control — a proportional gain compensator and a phase lead compensator — and compare their results with a physical tuned motor system. The design of these compensators must follow two criteria:

1. As low settling time T_s as possible.
2. A percent Overshoot of $\%OS \leq 10$.

In order to design either controller, we must use the design requirements to find necessary system parameters. Using the response overshoot requirement and the relationship between damping ratio and overshoot ($(\frac{\%OS}{100}) = \exp \left[-\frac{\pi \cdot \zeta}{\sqrt{1-\zeta^2}} \right]$), we can solve for the damping ratio:

$$\zeta = \sqrt{\frac{\ln(\frac{\%OS}{100})^2}{\ln(\frac{\%OS}{100})^2 + \pi^2}} = \sqrt{\frac{\ln(0.1)^2}{\ln(0.1)^2 + \pi^2}} = 0.591$$

Proportional Controller

The motor system block diagram is as follows:

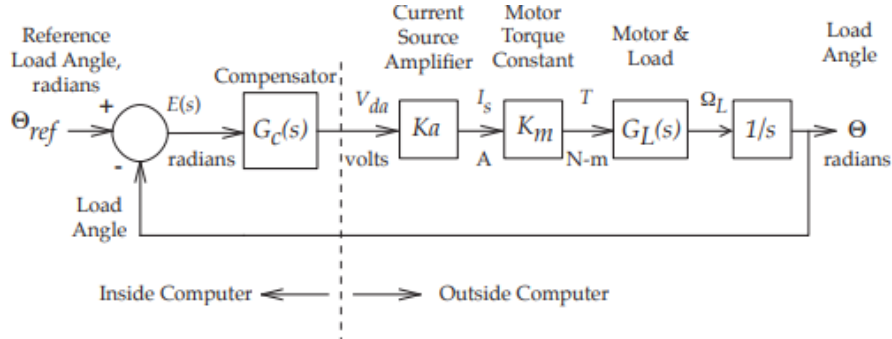


Figure 9: Motor Control Block Diagram

The proportional compensator with proportional Gain K_p is:

$$G_{C1}(s) = K_p.$$

To find the magnitude of the proportional gain, we can start by finding the characteristic equation of the unity feedback loop transfer function:

$$C.E. = 1 + K_p T_m(s) = 1 + K_p \frac{45.1}{s(0.34s + 1)} = 0 \quad \rightarrow \quad s^2 + \frac{1}{0.34}s + \frac{45.1}{0.34}K_p = 0$$

From here, we can use definitions of the natural frequency and damping ratio to solve for the required proportional gain:

$$\omega_n^2 = \frac{45.1}{0.34}K_p \quad \&\& \quad 2\zeta\omega_n = \frac{1}{0.34}$$

$$\rightarrow 4(0.591)^2 \left(\frac{45.1}{0.34} K_p \right) = \frac{1}{(0.34)^2} \rightarrow K_p = \frac{1}{4(0.591)^2(45.1)(0.34)} = \boxed{0.0467}$$

The gain chosen above is only valid if it meets the maximum allowable current, which occurs at $t = 0^+$ for a step reference Θ_0 :

$$i_{\max} = K_a K_p \Theta_0.$$

Rearranging this and taking into consideration $i_{\max} = 4.1$ A and $\Theta_0 = \pi$ rad, the upper bound on the proportional gain with respect to motor current capabilities is:

$$K_{p,cur} \leq \frac{i_{\max}}{K_a \Theta_0} = \frac{4.1}{0.41\pi} \approx 3.183 \frac{\text{V}}{\text{rad}}$$

Allowing a 10% buffer for the current drawn in the motor leaves us with a maximum gain value for the gain compensator as follows:

$$K_{p,allow} = 0.9 K_{p,cur} = 2.865 \frac{\text{V}}{\text{rad}}$$

This maximum value far surpasses the proportional gain value defined using the damping ratio, so $K_p = 0.0467$ may be kept.

Phase-Lead Compensator

The lead compensator has the form:

$$G_c(s) = \frac{K}{\beta} \frac{(s + \frac{1}{\tau})}{(s + \frac{1}{\tau\beta})}$$

The plant has its own phase margin PM_{T_m} . Both the plant and the compensator contribute to the overall phase margin, so the required maximum phase boost must be determined accordingly. A damping ratio of $\zeta = 0.591$ corresponds to a minimum phase margin requirement of $PM_{T_m} \geq 59^\circ$ for the compensator:

$$\phi_{\max} = 59^\circ - PM_{T_m} + 5^\circ$$

Note - a 5 degree correction factor was added for true system dynamics imperfections.

The phase angle and transfer function magnitude of the plant at a given frequency ω are:

$$\phi_{T_m}(j\omega) = -90^\circ - \arctan(0.34\omega) \qquad |T_m(j\omega)| = \frac{45.1}{\omega \sqrt{1 + (0.34\omega)^2}}$$

To determine the phase margin of the plant, the phase angle of the plant at the gain crossover frequency must be combined with the angle of the negative real axis in the complex plane. The gain crossover frequency is found by solving for the frequency at which the plant transfer

function magnitude equals unity:

$$|T_m(j\omega_{gc})| = \frac{45.1}{\omega_{gc}\sqrt{1 + (0.34\omega_{gc})^2}} = 1 \quad \rightarrow \quad 0.34^2\omega_{gc}^4 + \omega_{gc}^2 - 45.1^2 = 0$$

The gain crossover frequency is then obtained using the quadratic formula:

$$\omega_{gc}^2 = \frac{-1 \pm \sqrt{1 - 4(0.34^2)(-45.1^2)}}{2(0.34^2)} = 128.39, -139.04 \quad \rightarrow \quad \omega_{gc} = \sqrt{128.39} = 11.33$$

Using the gain crossover frequency, the phase margin of the plant and the required maximum phase boost are:

$$PM_{T_m} = 180^\circ + \phi_{T_m}(j\omega_{gc}) = 90^\circ - \arctan(0.34\omega_{gc}) = 90^\circ - \arctan(0.34 \cdot 11.33) = 14.55^\circ$$

$$\phi_{\max} = 59^\circ - 14.55^\circ + 5^\circ = 49.45^\circ$$

From the maximum phase boost, the non-dimensional parameter β for the pole and zero locations is calculated:

$$\beta = \frac{1 - \sin(\phi_{\max})}{1 + \sin(\phi_{\max})} = \frac{1 - \sin(49.45^\circ)}{1 + \sin(49.45^\circ)} = 0.1365$$

The compensator gain is determined from the current limit using the initial value theorem. Since the maximum current occurs at $t = 0^+$:

$$K_C = \frac{i_{\max}\beta}{K_a\Theta_0} = \frac{(4.1)(0.136)}{(0.41)(\pi)} = \boxed{0.4329}$$

To find the frequency at which the compensator provides its maximum phase boost, the uncompensated plant magnitude is multiplied by the compensator magnitude, which at $\omega_{\max}$ is $\frac{K_C}{\sqrt{\beta}}$:

$$\frac{45.1}{\omega_{\max}\sqrt{1 + (0.34\omega_{\max})^2}} \cdot \frac{K_C}{\sqrt{\beta}} = 1 \quad \rightarrow \quad 0 = 0.34^2\omega_{\max}^4 + \omega_{\max}^2 - 45.1^2 \frac{K_C^2}{\beta} \quad \rightarrow \quad \omega_{\max} = 12.30 \frac{\text{rad}}{\text{s}}$$

Using this frequency and β , the time constant τ is:

$$\tau = \frac{1}{\omega_{\max}\sqrt{\beta}} = \frac{1}{(12.30)\sqrt{0.136}} = 0.220 \text{ s}$$

Finally, with all parameters determined, the phase-lead compensator can be constructed by calculating the zero, pole, and overall gain:

$$z_c = \frac{1}{\tau} = \frac{1}{0.220} = 4.542 \quad p_c = \frac{1}{\beta\tau} = 33.28 \quad \frac{K_C}{\beta} = \frac{0.4329}{0.1365} = 3.172 \quad \boxed{G_C = 3.172 \frac{(s + 4.542)}{(s + 33.28)}}$$

4 Matlab Simulation Results

All responses are simulated to the same step reference angle of π radians.

4.1 Open-Loop Frequency Responses

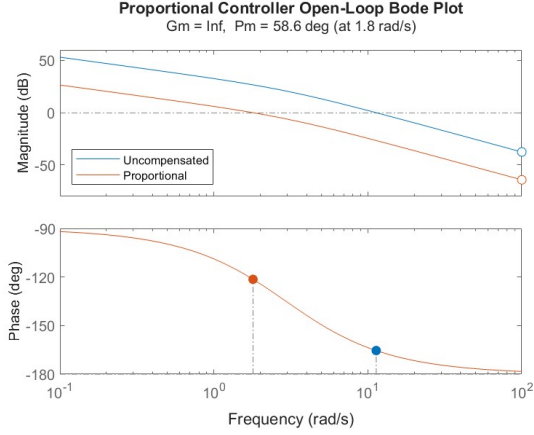


Figure 10: Proportional Open-Loop Bode Plot

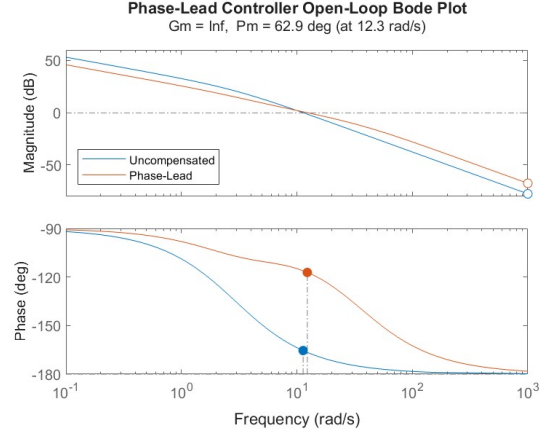


Figure 11: Phase-Lead Open-Loop Bode Plot

4.2 Closed-Loop Angle Step Response

Simulated response to a step of $\theta_{ref} = \pi$ rad. The overshoot and settling time recorded in MATLAB are listed in Appendix A (p. 10).

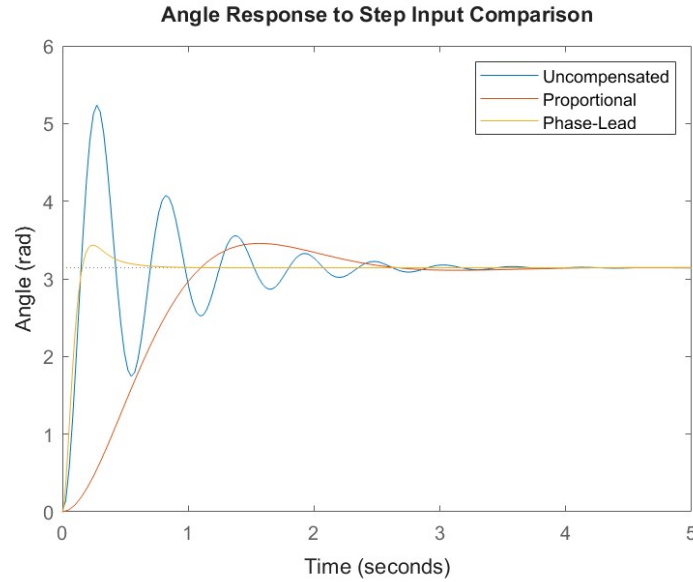


Figure 12: Proportional and Phase-Lead Closed-Loop Angle Responses to Step Input

4.3 Controller Output Voltage $V_{D/A}(t)$ Response

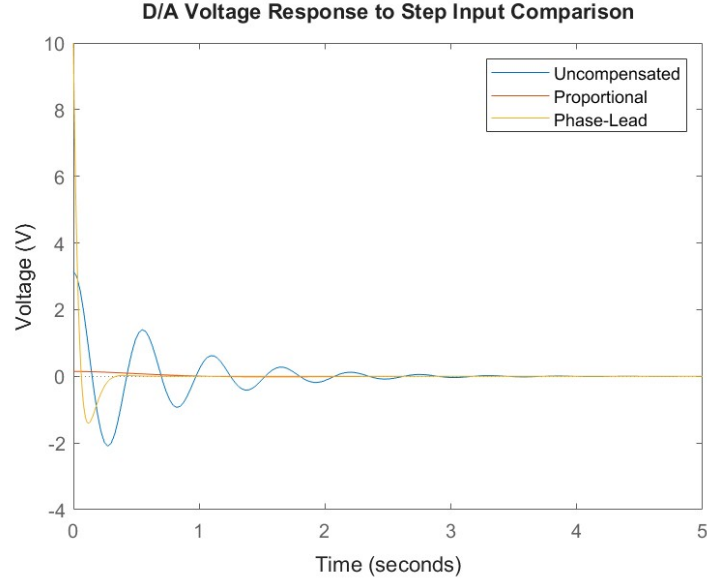


Figure 13: Proportional and Phase-Lead Closed-Loop Angle Responses to Step Input

Using the relationship between ideal current and D/A voltage $i(t) = K_a V_{D/A}(t)$ in conjunction with the maximum D/A voltage output values recorded from the step response plots, the maximum current for each compensated system can be found:

Proportional Compensator

$$i_{\max,P} = 0.06 \text{ A}$$

Phase Lead Compensator

$$i_{\max,C} = 4.1 \text{ A}$$

5 In-Class Experimental Results

5.1 Proportional Controller Response

Note: No plots are provided for the proportional compensator due to the lack of visible data.

The proportional compensator was tested more than 10 times. It was shown that around 50% of the time, the inertia did not move at all with the given step angle input $\Theta_0 = \pi$ radians. When the inertia did move, the current provided was still not enough to make it turn the full desired 180° , and the inertia only ended up turning a few degrees. Responses showed no movement in the inertia, and therefore no overshoot or settling time for the inertia. With a small "bump" input added by manually tapping the inertia as it received current, the inertia was able to move slightly further, but it never made a full 180° turn to its final desired output state.

5.2 Phase Lead Controller Response

The results of the phase-lead controlled response of the motor were much more conclusive than those of the proportional controller. The graphical output results of the phase-lead controller design are depicted below, with both the theoretical and experimental responses overlaid onto the same graph to show visually the differences between both outputs:

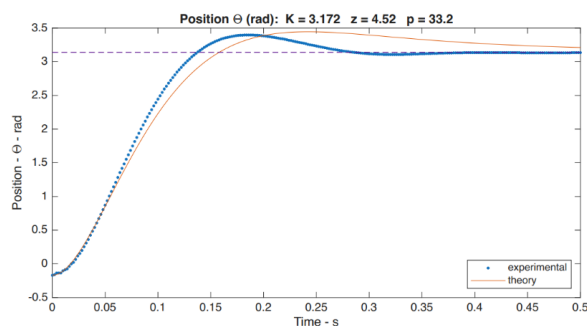


Figure 14: Angle Response to Step Angle Input Using Phase-Lead Controller Design

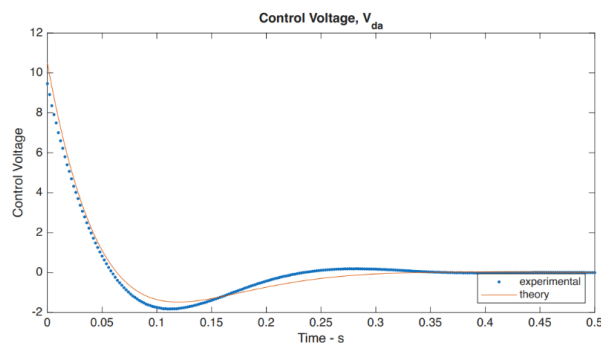


Figure 15: D/A Volt Response to Step Angle Input Using Phase-Lead Controller Design

In lab, the inertia was shown to turn very swiftly to its final desired 180° output with an input step. All trials done were visually identical; the inertia would immediately accelerate, passing its final angle with a slight overshoot and then immediately returning to rest close to exactly where it was expected to.

6 Discussion

6.1 Proportional Controller Analysis

The proportional compensator's theoretical overshoot is exactly 10%, and the theoretical settling time is 2.38 seconds. Given the inertia hardly turned at all during the experiment, it is expected that a higher proportional gain would lead to an increase in current drawing in the motor, which would likely increase the motor's ability to move.

The lack of movement in the experiment can be attributed to real-life factors and imperfections, including non-linear damping, non-ideal current, armature inductance and capacitance as well as shaft and inertia friction. The leading cause in this particular experiment is likely friction, as it directly increases the required force to accelerate the mass past its static friction, which means an increased supply current is needed.

Increasing the proportional gain to increase the supplied current to move the inertia is not a perfect solution, as increasing the gain directly decreases the closed-loop response stability and therefore both the settling time and overshoot. The inertia would be supplied with enough current to overcome static friction, but this increase would also provide it with enough to far surpass its output angle goal, causing oscillation around the target angle and therefore increasing the settling time.

This is shown theoretically too, as the closed-loop poles would decrease in magnitude with an increase in open-loop gain added to the system:

$$\text{Closed-Loop Characteristic Equation: } C.E. = P(s) + K_p Z(s)$$

Where $Z(s)$ is the numerator of the open-loop transfer function and $P(s)$ is the denominator, representing zeroes and poles respectively. In the original uncompensated system, there are no zeroes, so the gain contributes solely to the zero-degree term in the characteristic equation, and therefore exclusively contributes to the leftward movement of the pole locations. As a decrease in magnitude of pole values increases their dominance, it also decreases their stability as they move closer to the imaginary axis in the s-plane, decreasing damping and therefore increasing oscillations.

The proportional controller is a great demonstration of why theory can only give so much information — the chosen gain was theoretically valid but physically insufficient and needed to be adjusted for the physical system even though it worked very well in the model.

6.2 Phase-Lead Controller Analysis

The overall overlap between the theoretical and experimental responses is very high — their responses are very similar and follow the same dynamics and oscillations. The primary difference to highlight is present in the output angle response. The experimental response appears to have much tighter oscillations and settles much faster at the desired output angle than the theoretical response. Although this is generally a good outcome, the decreased settling time can be directly attributed to system imperfections, notably friction — the system oscillates less far than the theoretical model because friction decreases the oscillatory acceleration of the inertia as it passes its final value.

Aside from the difference in the output angle response settling time, there is no major difference - both the model and the experiment overshoot visually the same amount before settling to the final 180 °angle, meaning the actual model overshoots the predicted 9.3%. Strong initial acceleration and inertial components likely overcome any imperfections in the system's functionality, including friction.

The D/A voltage responses are visually quite similar, with the only exception being the amplitude of oscillations — the experimental results showed a very small increase in amplitude of oscillations, likely of around 5-10%. This is likely due to the same reasons as listed prior, but may also be attributed to friction and damping nonlinearities, decreasing their overall contribution when the inertia has a lower angular velocity.

7 Conclusion

This laboratory demonstrated the design and implementation of proportional and phase-lead controllers for DC motor position control. The proportional controller, while theoretically meeting overshoot and settling time requirements, proved insufficient in practice due to friction and current limitations. In contrast, the phase-lead compensator achieved fast settling and acceptable overshoot, closely matching simulation predictions. Minor discrepancies in D/A voltage oscillations highlight the influence of unmodeled dynamics and noise. Overall, the experiment validated the importance of advanced compensator design and highlighted the gap between ideal models and real-world systems.

8 Appendix

Matlab Output Values and Calculation Results

```
--- Proportional Controller ---
K_p = 0.0467
Phase Margin = 113.44 deg at Inf rad/s

--- Phase-Lead Compensator ---
phi_max = 49.45 deg
beta = 0.1365
K_c = 0.4344
tau = 0.220 s
Zero = 4.550 rad/s
Pole = 33.342 rad/s
Gain = 3.183
Phase Margin = 128.66 deg at Inf rad/s

--- Proportional Controller: Angle Response Design Criteria ---
Percent Overshoot = 10.00
Settling Time = 2.38 s

--- Phase-Lead Compensator: Angle Response Design Criteria ---
Percent Overshoot = 9.28
Settling Time = 0.51 s

--- Proportional Controller: Maximum Current ---
Maximum Current = 0.06
Maximum Location = 0.00

--- Phase-Lead Compensator: Maximum Current ---
Maximum Current = 4.10
Maximum Location = 0.00
```

Figure 16: Proportional and Phase-Lead Closed-Loop Angle Responses to Step Input

Matlab Code

Full script used for design and simulation:

```
1 clear;
2 clc;
3 figs = findall(0, 'Type', 'figure');
4 for k = 1:length(figs)
5     clf(figs(k));
6 end
7
8 %% Variable Definitions %%
9 K_a = 0.41; % A/V
10 K_m = 0.11; % N*m/A
```

```

11 B = 0.001; % N*m*s/rad
12 J_tot = 0.00034; % N*m*s^2/rad
13 i_max = 4.1; % A
14 theta_0 = pi; % rad
15
16 K_da = K_a*K_m / B;
17 tau_m = J_tot / B;
18
19 OS = 0.1;
20 PM_pl = 59; % deg
21 zeta = sqrt(((log(OS))^2) / ((log(OS))^2 + pi^2));
22
23 %% Plant Definition %%
24 % Plant: Tm(s) = 45.1 / (s(0.34s + 1))
25 G = tf(K_da, [tau_m 1 0]);
26 % Closed-loop uncompensated
27 T_m = feedback(G, 1);
28
29 %% Proportional Controller %%
30 K_p = ((2*zeta)^2 * K_da * tau_m)^(-1);
31 C_p = zpk([], [], K_p);
32 G_p = series(G, C_p);
33
34 % Closed-loop with proportional controller
35 T_p = feedback(G_p, 1);
36 [GM_p, PM_p, w_gc_p, w_pc_p] = margin(T_p);
37
38 % Upper bound proportional controller gain
39 K_pmax = (i_max) / (K_a * theta_0);
40 K_pmax_buffer = 0.9*K_pmax;
41
42 %% Phase-Lead Compensator Design %%
43 % Values derived analytically
44 w_gc = sqrt((-1 + sqrt(1 - 4*(tau_m^2)*(-K_da^2))) / (2*(tau_m^2)));
45 PM_Tm = 180 + (-90 - atand(tau_m * w_gc));
46 phi_max = PM_pl - PM_Tm + 5;
47 beta = (1 - sind(phi_max)) / (1 + sind(phi_max));
48 K_c = (i_max*beta) / (K_a*theta_0); % from current limit
49 w_max = sqrt((-1 + sqrt(1 - 4*(tau_m^2)*(-(K_da*K_c)^2) / beta))) /
    (2*(tau_m^2)); % rad/s
50 tau = 1 / (w_max * sqrt(beta));
51
52 % Build compensator
53 C_lead = zpk([-1/tau], [-1/(beta*tau)], K_c/beta);
54 G_c = series(G, C_lead);
55
56 % Closed-loop with lead compensator

```

```

57 T_c = feedback(G_c, 1);
58 [GM_c, PM_c, w_gc_c, w_pc_c] = margin(T_c);
59
60 %% Display Final Values %%
61 disp('--- Proportional Controller ---')
62 fprintf('K_p = %.4f\n', K_p);
63 fprintf('Phase Margin = %.2f deg at %.2f rad/s\n', PM_p, w_gc_p);
64
65 disp(' ')
66 disp('--- Phase-Lead Compensator ---')
67 fprintf('phi_max = %.2f deg\n', phi_max);
68 fprintf('beta = %.4f\n', beta);
69 fprintf('K_c = %.4f\n', K_c);
70 fprintf('tau = %.3f s\n', tau);
71 fprintf('Zero = %.3f rad/s\n', 1/tau);
72 fprintf('Pole = %.3f rad/s\n', 1/(beta*tau));
73 fprintf('Gain = %.3f\n', K_c/beta);
74 fprintf('Phase Margin = %.2f deg at %.2f rad/s\n', PM_c, w_gc_c);
75
76 %% Bode Plots %%
77 figure(1);
78 margin(G);
79 hold on;
80 margin(G_p);
81 hold off;
82 title('Proportional Controller Open-Loop Bode Plot');
83 legend('Uncompensated', 'Proportional', 'Location', 'southwest');
84
85 figure(2);
86 margin(G);
87 hold on;
88 margin(G_c);
89 hold off;
90 title('Phase-Lead Controller Open-Loop Bode Plot');
91 legend('Uncompensated', 'Phase-Lead', 'Location', 'southwest');
92
93 %% Step Responses %%
94 figure(3);
95 step(theta_0*T_m, theta_0*T_p, theta_0*T_c, 5);
96 legend('Uncompensated', 'Proportional', 'Phase-Lead');
97 title('Angle Response to Step Input Comparison');
98 ylabel("Angle (rad)");
99
100 figure(4);
101 step(theta_0*feedback(1, G), theta_0*feedback(C_p, G), theta_0*
    feedback(C_lead, G), 5);
102 legend('Uncompensated', 'Proportional', 'Phase-Lead');

```

```

103 title('D/A Voltage Response to Step Input Comparison');
104 ylabel("Voltage (V)");
105
106 %% Display Final Step Response Metrics %%
107 P_Ang_Info = stepinfo(T_p);
108 C_Ang_Info = stepinfo(theta_0*T_c);
109 P_Volt_Info = stepinfo(theta_0*feedback(C_p, G));
110 C_Volt_Info = stepinfo(theta_0*feedback(C_lead, G));
111
112 disp(' ')
113 disp('--- Proportional Controller: Angle Response Design Criteria
    ---')
114 fprintf("Percent Overshoot = %.2f \n", P_Ang_Info.Overshoot)
115 fprintf("Settling Time = %.2f s \n", P_Ang_Info.SettlingTime)
116
117 disp(' ')
118 disp('--- Phase-Lead Compensator: Angle Response Design Criteria ---
    ')
119 fprintf("Percent Overshoot = %.2f \n", C_Ang_Info.Overshoot)
120 fprintf("Settling Time = %.2f s \n", C_Ang_Info.SettlingTime)
121
122 disp(' ')
123 disp('--- Proportional Controller: Maximum Current ---')
124 fprintf("Maximum Current = %.2f \n", K_a*P_Volt_Info.Peak)
125 fprintf("Maximum Location = %.2f \n", K_a*P_Volt_Info.PeakTime)
126
127 disp(' ')
128 disp('--- Phase-Lead Compensator: Maximum Current ---')
129 fprintf("Maximum Current = %.2f \n", K_a*C_Volt_Info.Peak)
130 fprintf("Maximum Location = %.2f \n", K_a*C_Volt_Info.PeakTime)

```